

TABLE III
PCB Prototyping Machines
from Different Manufacturers

Manufacturer	Product	System
MitsPCB	FP21T	Mechanical
AccurateCNC	Accurate 426	Mechanical
Colinbus	CPR series , PCB box	Mechanical
T-Tech	QC series	Mechanical
LPKF Laser and Electronics	ProtoMat S103	Laser
Everprecision Tech Co	EP-2006	Laser

TABLE IV
SMT Pick-and-Place Machines

Manufacturer	Products
LPKF Laser and Electronics	Protoplace S
Mechatronics Systems	M1, P1
SMTnet	Manncorp 7700-FV Auto Placer, Manncorp SMT 2000 (manual)
Essemtec	Paraquda, Pantera-X, Cobra

consuming. Besides, making double-layer PCBs with such techniques is very cumbersome.

Various desktop PCB-prototyping systems available these days help in making the PCB prototypes very quickly (refer Fig. 2). These systems are majorly of two types: mechanical and laser milling systems. Both types of systems create conductive paths and pads by milling insulating paths. The insulating paths separate the electro-conductive copper surfaces that form the network of conductive paths.

A mechanical milling system contains a rotating drill head that works on the copper-clad sheet, milling all insulating paths. It is comparatively slower than a laser system which uses



Fig. 2: A PCB-prototyping machine



Fig. 3: A desktop SMT pick-and-place machine



Fig. 4: A reflow oven

an energy-emission device to focus a highly concentrated stream of photons onto a small area of a work-piece for milling the insulating paths. Table III shows some mechanical and laser-type PCB-prototyping systems from different manufacturers.

Next, all the required holes need to be drilled. Both these systems can drill holes for mounting through-hole components. The specifications of these systems are simple and easy to understand. You can look at specs, such as working area, resolution, speed, drill sizes and power consumption, to decide a suitable system for you.

SMT pick and place

Once the PCBs are ready, you need to mount all the components and test the boards. Mounting of through-hole components is easy but SMD components require some skill and patience. So most people avoid doing it themselves and outsource the job, which costs money for making stencils and tooling, and time delay is always there.

With desktop SMT machines the SMD boards can be assembled in-house. Fig. 3 shows a desktop SMT pick-and-place machine. Such machines come in completely automatic and semi-automatic forms.

In an automatic system, no operator intervention is required. The system automatically applies solder paste on the pad and then places the components on corresponding solder beds. The placement is not very accurate in most cases, but during reflow oven stage the parts tend to organise nicely.

In a manual system, the operator controls the application of solder paste and placement of parts. The placement can be much more accurate in this case. But you may not want an operator, who could make errors. Besides, the process will be much slower than with an automatic machine.

Some manufacturers have also come up with solutions that include PCB milling, solder-paste application and pick and place, all in one system.



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